CS310 Final Exam S 1 2022 - Solutions

Wireless Networks (The University of the South Pacific)

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Student Name: _____ Student Number:_____ Seat: _____

Possible Solutions

CS310: Computer Networks

School of Information Technology, Engineering, Mathematics, and Physics

Final Examination
Semester 1 2022

F2F/Remote

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 3 hours

Total marks: 100

Instructions:

1. *This exam is a closed book exam.*
2. *This exam has three sections:*
   - Section A: 30 Marks
   - Section B: 40 Marks
   - Section C: 30 Marks
3. *This exam has 11 questions and ALL are compulsory.*
4. *There are 14 pages in this paper including this cover page.*
5. *Show all necessary working. Partial marks will be awarded for partially correct answers.*
6. *Only non-programmable calculators are allowed.*
7. *This exam is worth 40 % of your overall mark.*
8. *The minimum passing mark is 40/100.*

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Section A (Questions 1) - Short and Long Answer Questions (30 Marks)

a) What is the difference between a host and an end system? List several different types of end systems. Is a Web server an end system? (4 marks)

There is no difference. Throughout this text, the words "host" and "end system" are used interchangeably. End systems include PCs, workstations, Web servers, mail servers, PDAs, Internet-connected game consoles, etc.

b) Consider the following two 16-bit words (in binary). Remember that we compute the one's complement sum [1] of the two words to compute the Internet checksum of a set of 16-bit words. That is, we add the two integers together, ensuring that any carry into the 17th bit of the initial total is returned to the 1's position of the final sum; we then take the one's complement of the result. Compute the Internet checksum value for these two 16-bit words:

10110101 01101111 this binary number is 46447 decimal (base 10)

00011101 11100111 this binary number is 7655 decimal (base 10)

What is the sum of these two 16 bit numbers? Don't put any spaces in your answer.

(5 marks)

*Hint: Show the sum and the internet checksum is the one's complement of the sum.

1. The sum of 10110101 01101111 and 00011101 11100111 = 11010011 01010110

2. The internet checksum is the one's complement of the sum:
11010011 01010110 = 00101100 10101001

Markers Note: 2.5 marks for each correct point.

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c) The table below lists the MTUs for many different protocols. The MTUs range from 296 to 65,535.

Table 1: MTUs for many different protocols

Protocol	MTU
Hyperchannel	65,535
Token Ring (16 Mbps)	17, 914
Token Ring (4 Mbps)	4, 464
FDDI	4, 352
Ethernet	1, 500
X.25	576
PPP	296

i. What does MTU stands for? (1 marks)

MTU stands for the **maximum transmission unit**.

ii. What would be an advantage of having a large MTU as in the table 1.? (2 marks)

A larger MTU brings **greater efficiency in transmitting**.

[Because each network packet carries more user data with protocol overheads, such as headers or underlying per-packet delays, remain fixed; the resulting higher efficiency means an improvement in bulk protocol throughput.]

iii. What would be the advantage of having a small MTU as in the table 1? (2 marks)

Smaller MTU values can **reduce network delay**.

d) Imagine that you are trying to visit www.enterprise.com, but you don't remember the IP address the web-server is running on.

Assume the following records are on the TLD DNS server:

- (www.enterprise.com, dns.enterprise.com, NS)
- (dns.enterprise.com, 146.54.200.182, A)

Assume the following records are on the enterprise.com DNS server:

- (www.enterprise.com, west3.enterprise.com, CNAME)
- (west3.enterprise.com, 142.81.17.206, A)
- (enterprise.com, mail.enterprise.com, MX)
- (mail.enterprise.com, 247.29.133.173, A)

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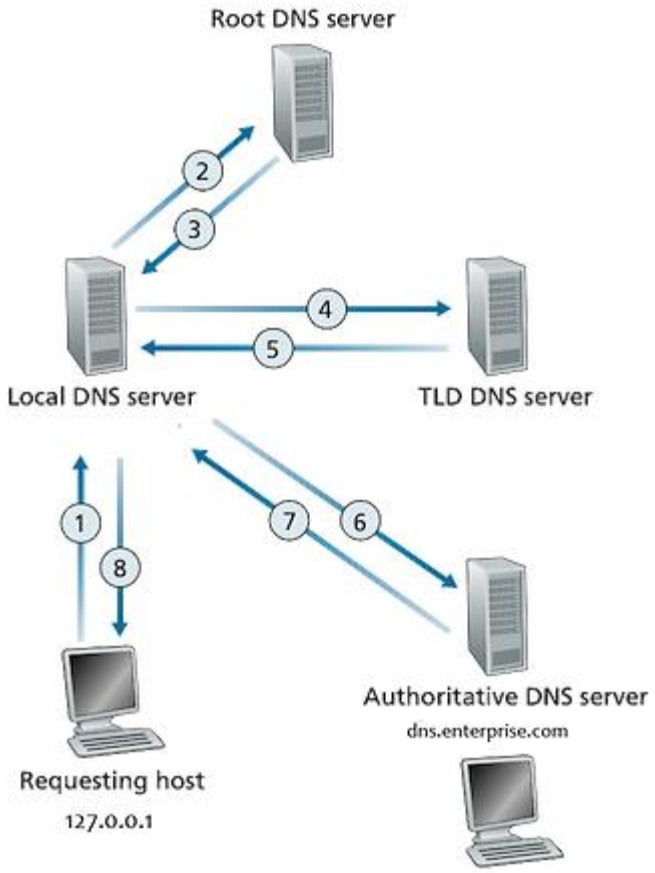

Figure 1: A Senario on DNS concepts.

Assume your local DNS server only has the TLD DNS server cached. What transport protocol(s) does DNS use: TCP, UDP, or Both? (2 marks)

_____

_____

e) Consider an 8-bit host address datagram network. Suppose a router uses longest-prefix matching, and has the following forwarding table:

Table 2: The sample of the longest prefix matching.

Prefix Match	Interface
01	1
11	2

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011	3
111	4
100	5
Otherwise	6

Suppose a datagram arrives at the router, with destination address 00111001. To which interface will this datagram be forwarded using longest-prefix matching? (4 marks)

1.Since the address is 00111001, it will go to interface 6.

2. Since the address is 01101001, it will go to interface 3.

3. Since the address is 00110100, it will go to interface 6.

f) The figure below shows the format of an IP datagram to answer from (i to iv).

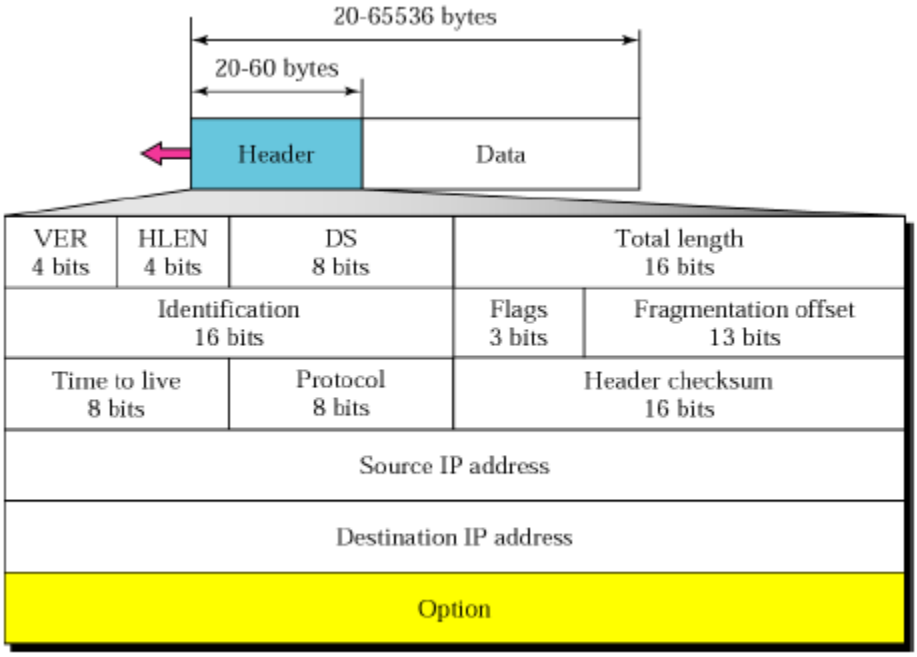

Figure 2: The format of an IP datagram.

i. Which version of the IP protocol is being used in this transmission? (1 mark)

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Version 4 (IPv4)

ii. Is the packet fragmented? If yes, what is the offset? (3 marks)

Yes, Fragment Offset field > 0 then it is a packet fragment. = (13 bits)

iii. What is the total length of the IP datagram? (3 marks)

This is a 16-bit field that defines the total length(Header + data) of the IP datagram in bytes. Since the field is 16 bits, the total length of the IP datagram is limited to 2^16 = 65536 bytes, of which 20-40 bytes are the header add the rest is data.

iv. Describe the understanding of fragmentation process? (3 Marks)

Fragmentation is a process of breaking down the IP packets into smaller pieces(fragments). Fragmentation is required when the datagram is larger than the MTU. Each fragment then becomes a datagram in itself and transmitted independently from source..These datagrams are reassembled by the destination.

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Section B (Questions 2) - Short and Long Answer Questions (40 Marks)

a) In the scenario below, the left and right clients communicate with a server using UDP sockets. The same socket at the server is used to communicate with both clients. The Python code used to create the sockets is shown in the figure. Consider the four transport-layer packets – A, B, C and D – shown in the figure below.

(8 marks)

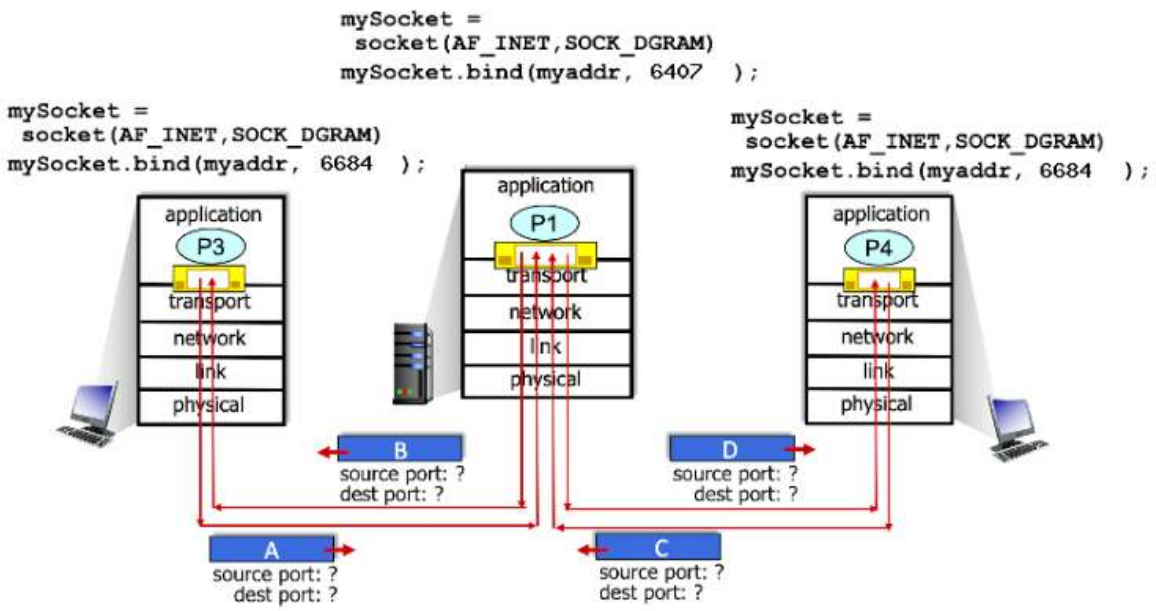

Figure 3: Portayal of the four transport-layer packets – A, B, C and D.

1. The source port for packet D is port 6407.

2. The destination port for packet D is port 6684.

3. The source port for packet C is port 6684.

4. The destination port for packet C is port 6407.

5. The source port for packet B is port 6407.

6. The destination port for packet B is port 6684.

7. The source port for packet A is port 6684.

8. The destination port for packet A is port 6407.

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b) Consider the 6-node network shown below, with the given link costs.

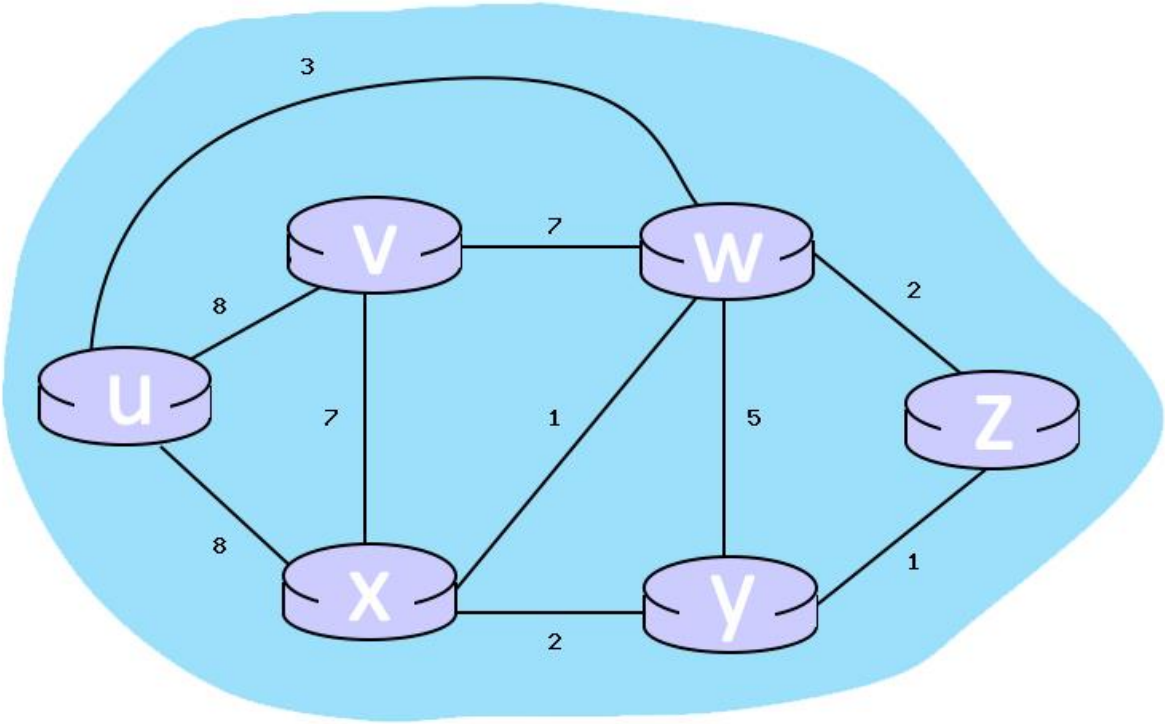

Figure 4: Portayal of the 6-node network.

Using Dijkstra's algorithm, find the least cost path from source node U to all other destinations and answer the following questions. (10 marks)

1. The minimum distance from node u to node z is 5, and node z's predecessor is node w. The full answer was: 5,w

2. The minimum distance from node u to node v is 8, and node v's predecessor is node u. The full answer was: 8,u

3. The minimum distance from node u to node w is 3, and node w's predecessor is node u. The full answer was: 3,u

c) <u>TCP Retransmissions (Reliable Data Transmission With Ack Loss)</u>

Consider the figure below in which a TCP sender and receiver communicate over a connection in which the segments can be lost. The TCP sender wants to send a total of 10 segments to the receiver and sends an initial window of 5 segments at t = 1, 2, 3, 4,

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and 5, respectively. Suppose the initial value of the sequence number is 88 and every segment sent to the receiver each contains 775 bytes. The delay between the sender and receiver is 7 time units, and so the first segment arrives at the receiver at t = 8, and an ACK for this segment arrives at t = 15. As shown in the figure, 3 of the 5 segments is lost between the sender and the receiver, but none of the ACKs are lost. Assume there are no timeouts and any out of order segments received are thrown out. (16 marks)

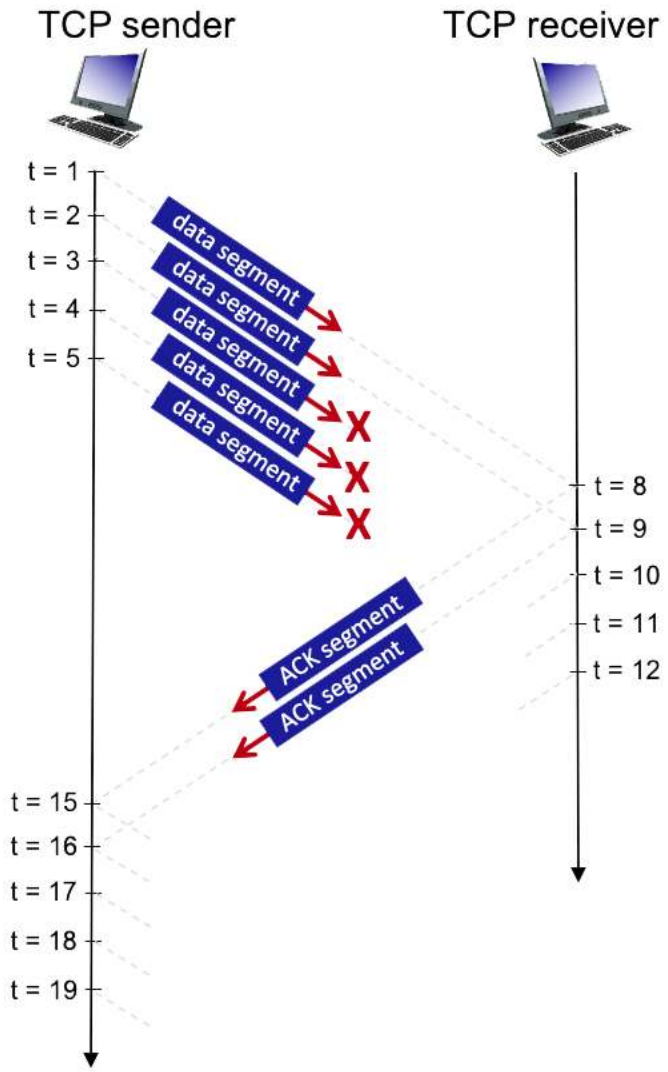

Figure 5: Portayal of the TCP Retransmissions.

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1. The sequence number of the first segment is the starting sequence number, which is 88.

2. The sequence number of the second segment is = 88 + 775 = 863.

3. The sequence number of the third segment is = 863 + 775 = 1638.

4. The sequence number of the fourth segment is = 1638 + 775 = 2413.

5. The sequence number of the fifth segment is = 2413 + 775 = 3188.

6. The ACK value is the sequence number of the next expected segment, which is 863.

7. The ACK value is the sequence number of the next expected segment, which is 1638.

8. Since the segment was lost, the ACK is never sent, so the answer is x.

9. Since the segment was lost, the ACK is never sent, so the answer is x.

10. Since the segment was lost, the ACK is never sent, so the answer is x.

11. The sequence number of this segment is calculated the same way as the first five segments. The sequence number is 3963.

12. The sequence number of this segment is calculated the same way as the first five segments. The sequence number is 4738.

13. Since there was nothing to send (as we're waiting for ACKs), the answer is x.

14. Since there was nothing to send (as we're waiting for ACKs), the answer is x.

15. Since there was nothing to send (as we're waiting for ACKs), the answer is x

d) Briefly describe what are the functions of Transport, Network and Data Link layer. (6 marks)

Direct from lecture note: thus 2 marks each

Section C (Questions 3) - Short and Long Answer Questions (30 Marks)

The University of South Pacific is building a new Engineering building at the main campus. There will be 2 departments based in the engineering building. The number of IP addresses required on a per department basis is as follows:

- Electrical Engineering – 120 IP Addresses
- Electronics Engineering – 45 IP Addresses

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Part A (20 marks)

a. Subnet the major network 144.120.0.0/16 using Variable Length Subnet Mask to design the network for the engineering building.
Fill the table below. All the calculations should be clearly stated.

Table 3: Carries the subnet informations.

Department	Network	Usable IP Range	Broadcast Address
Electrical			
Electronics			

Solution – Part A

Step 1 - [2 Marks]

- 144.120.0.0/16
- 144 – First octet is between 128 – 191, The address is Class B.

Step 2 [2 Marks]

- Default Subnet Mask for Class B is 255.255.0.0
- Default Subnet Mask for Class B in Binary is:
  - 11111111.11111111.00000000.00000000

Step 3 [1 Mark]

- Identify the Largest Network
  - Electrical Engineering with 120 IP Addresses
- To accommodate 120 IP Address
  - Usable IP Address = 2^H-2

H = 7, since 2^8-2 = 126.

We require 120 Addresses and the 126 is supplied.

Step 4 [2 Marks]

- Identify the New Subnet Mask
- Default Subnet Mask : 11111111.11111111.00000000.00000000
- Reserve 7 Bits from the Right and Turn the Remaining bits into 1's.
- New Subnet Mask is : 11111111.11111111.11111111.10000000
- New Subnet Mask in Decimal : 255.255.255.128
- New Subnet Mask in CIDR : /25

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Step 5 [2 Marks]

- Calculate the Update
  - Update = 256 – Last Octet of Change
    - 256 -128 = 128

Step 6 [1 Mark]

- Carry out the Network Listing

Network	**Usable IP**	**Broadcast**
144.120.0.0/25	144.120.0.1-144.120.0.126	144.120.0.127
144.120.0.128/25	144.120.0.129-144.120.0.254	144.120.0.255
144.120.1.0/25	144.120.1.1-144.120.1.126	144.120.1.126

Step 7 [1 Mark]

Thus the first network 144.120.0.0/25 can be allocated to the Electrical Department.

Step 8 – Consider the 2nd Largest network. [1 Mark]

The second largest network is Electronics Engineering with 45 IP address requirements.

- 144.120.0.128/25

Step 9 [1 Mark]

- /25 is 255.255.255.128
- /25 in Binary is
  - 11111111.11111111.11111111.10000000

Step 10 [1 Mark]

- To accommodate 45 IP Address
  - Usable IP Address = 2^H-2

    H = 6, since 2^6-2 = 62.

  We require 45 Addresses and the 62 is supplied.

Step 11 [2 Marks]

- Identify the New Subnet Mask
- Existing Subnet Mask : 11111111.11111111.11111111.10000000
- Reserve 6 Bits from the Right and Turn the Remaining bits into 1's.
- New Subnet Mask is : 11111111.11111111.11111111.11000000
- New Subnet Mask in Decimal : 255.255.255.192
- New Subnet Mask in CIDR : /26

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Step 12 [1 Mark]

- Calculate the Update
  - Update = 256 – Last Octet of Change
    - 256 -192 = 64

Step 13 [1 Mark]

- Carry out the Network Listing

Network	Usable IP	Broadcast
144.120.0.128/26	144.120.0.129-144.120.0.190	144.120.0.191
144.120.0.192/26	144.120.0.193-144.120.0.254	144.120.0.255
144.120.1.0/26	144.120.1.1-144.120.1.62	144.120.1.63

Step 14 [1 Mark]

The Network 144.120.0.128/26 can be allocated to the Electronics Department.

Step 15 – The table will be as follows [1 Mark]

Department	**Network**	**Usable IP Range**	**Broadcast Address**
Electrical	144.120.0.0/25	144.120.0.1-144.120.0.126	144.120.0.127
Electronics	144.120.0.128/26	144.120.0.129-144.120.0.190	144.120.0.191

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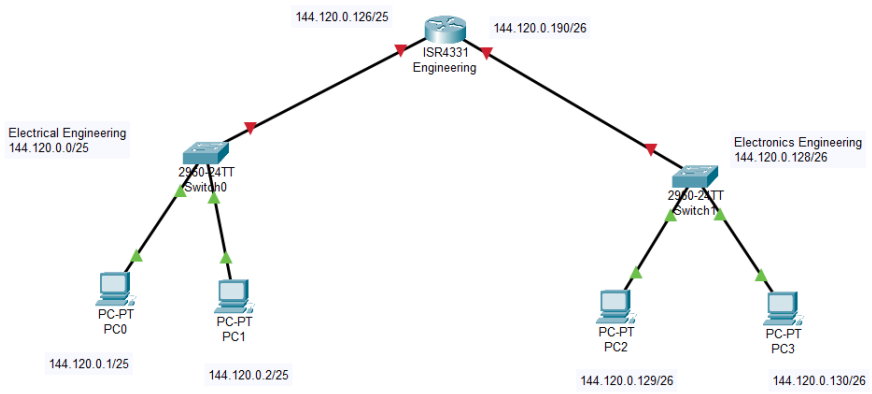
Part B

a. Draw the topology diagram. Use the last Usable IP address as for the Default gateway.

The topology diagram is as follows: (7 marks)

Solution Part B

The topology diagram is as follows [2 Mark]



Part C

Explain Network Address Translation. (3 marks)

Solution Part C

Network Address Translation

Network Address Translation is the process of mapping the private IP space into a public IP address or addresses. Example the internal network would operate on the IP space 10.0.0.0/8 whilst for internet access, the entire 10.0.0.0/8 space can be mapped to a public address 202.62.100.100

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THE END

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